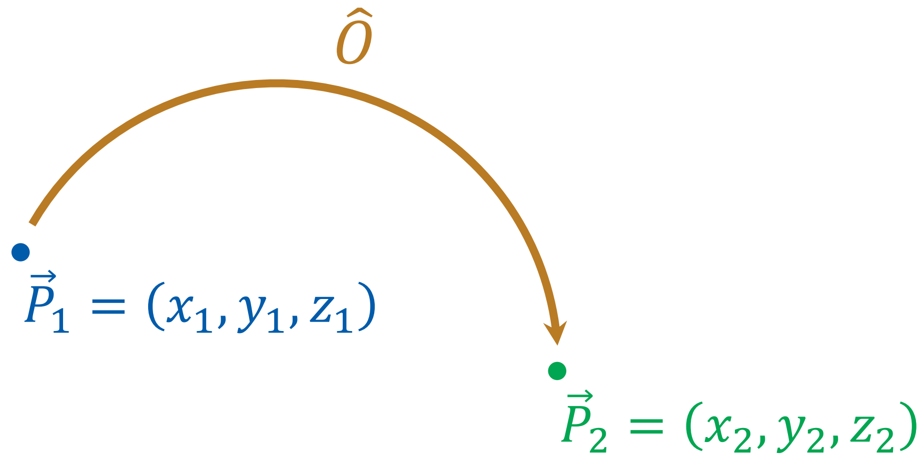
화학수학

# 열네 번째 수업

## 행렬

# Symmetry Operators




$$\hat{O}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\hat{O}\vec{P}_1 = \vec{P}_2$$

# Point Symmetry Operators

## (점대칭 연산자)

a class of symmetry operators
that leave at least one point unmoved
(the unmoved point = origin)

Identity operator $\hat{E}$

Inversion operator $\hat{\imath}$

Reflection operator $\hat{\sigma}$

Rotation operator $\hat{C}_n, \hat{S}_n$

# Symmetry Elements (대칭 요소)

Point, line, or plane relative to which
the symmetry operation is performed

- Must include the origin
- Sometimes denoted by the same symbol

- Inversion $i$: the origin
- Reflection $\sigma$: the reflection plane
- Rotation $C_n$, $S_n$: the rotation axis

# Symmetry of Object

- If after a symmetry operation, all particles in the object is in the same conformation as before, the symmetry operator belongs to the object.

- Coordinate system
  - origin: center of mass of the object
  - z axis: the rotation axis of highest order (largest $n$)

# Symmetry Operators on Functions

If an operator $\hat{O}$ operates as $\hat{O}(x_1, y_1, z_1) = (x_2, y_2, z_2)$, for a function $f$,

$$\hat{O}f = g$$

where $g(x_2, y_2, z_2) = f(x_1, y_1, z_1)$.

Use the relation between $(x_1, y_1, z_1)$ and $(x_2, y_2, z_2)$!

# Eigenfunctions of Symmetry Operators

- Eigenvalue = +1

  the operator leaves a function unchanged

- Eigenvalue = -1

  the operator changes the sign of the function's value at all points

# 오늘의 수업

- Matrices
  - Matrix and matrix algebra
  - Special matrices
  - Partners of a matrix
  - Properties of a matrix

# Matrix (행렬)

A list of quantities

arranged in rows (행) and columns (열)

matrix elements
(행렬 원소)

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

($m$ by $n$ matrix)

# Some Special Matrices

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

- $m = n$: square matrix (정사각행렬)
- $m = 1$: row vector (행벡터)
- $n = 1$: column vector (열벡터)

# More on Square Matrices (1)

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix}$$

off-diagonal elements
(비대각원소)

diagonal elements
(대각원소)

# More on Square Matrices (2)

- Diagonal matrix (대각행렬): off-diagonal elements = 0

$$\mathbf{A} = \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}$$

- Upper triangular matrix (위삼각행렬)

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

- Lower triangular matrix (아래삼각행렬)

$$\mathbf{A} = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

# Matrix Algebra

- Two matrices are equal if and only if (1) they have the same numbers of rows and columns and (2) every element of one matrix is equal to the corresponding element of the other matrix
- Sum of two matrices: $\mathbf{C} = \mathbf{A} + \mathbf{B}$ if and only if (iff)

$$c_{ij} = a_{ij} + b_{ij} \text{ for every } i \text{ and } j$$

- Product of a scalar $c$ and a matrix $\mathbf{A}$: $\mathbf{B} = c\mathbf{A}$ iff

$$b_{ij} = ca_{ij} \text{ for every } i \text{ and } j$$

# Product of Two Matrices

- Matrix product $\mathbf{C} = \mathbf{AB}$ iff

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

where $n$ is the number of columns in $\mathbf{A}$ and the number of rows in $\mathbf{B}$.

($m$ by $n$ matrix) × ($n$ by $p$ matrix) = ($m$ by $p$ matrix)

# Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \text{ x } \begin{bmatrix} 10 & 11 \\ 20 & 21 \\ 30 & 31 \end{bmatrix}$$

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

$$= \begin{bmatrix} 1\text{x}10 + 2\text{x}20 + 3\text{x}30 & 1\text{x}11 + 2\text{x}21 + 3\text{x}31 \\ 4\text{x}10 + 5\text{x}20 + 6\text{x}30 & 4\text{x}11 + 5\text{x}21 + 6\text{x}31 \end{bmatrix}$$

$$= \begin{bmatrix} 10+40+90 & 11+42+93 \\ 40+100+180 & 44+105+186 \end{bmatrix} = \begin{bmatrix} 140 & 146 \\ 320 & 335 \end{bmatrix}$$

(Image: https://towardsdatascience.com/a-complete-beginners-guide-to-matrix-multiplication-for-data-science-with-python-numpy-9274ecfc1dc6)

# Example 13.15

Find the matrix product

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 2 \\ 3 & 0 & 1 \\ 1 & 1 & -1 \end{bmatrix}.$$

The matrix product is

$$\begin{bmatrix} 1 \times 0 + 0 \times 3 + 2 \times 1 & 1 \times 0 + 0 \times 0 + 2 \times 1 & 1 \times 2 + 0 \times 1 + 2 \times (-1) \\ 0 \times 0 + (-1) \times 3 + 1 \times 1 & 0 \times 0 + (-1) \times 0 + 1 \times 1 & 0 \times 2 + (-1) \times 1 + 1 \times (-1) \\ 0 \times 0 + 0 \times 3 + 1 \times 1 & 0 \times 0 + 0 \times 0 + 1 \times 1 & 0 \times 2 + 0 \times 1 + 1 \times (-1) \end{bmatrix}$$

which is

$$\begin{bmatrix} 2 & 2 & 0 \\ -2 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

# Scalar Product of Vectors

For two vectors $\vec{F} = \left(F_x, F_y, F_z\right)$ and $\vec{G} = \left(G_x, G_y, G_z\right)$,

their scalar product can be seen as

$$\vec{F} \cdot \vec{G} = F_x G_x + F_y G_y + F_z G_z = \begin{bmatrix} F_x & F_y & F_z \end{bmatrix} \begin{bmatrix} G_x \\ G_y \\ G_z \end{bmatrix}$$

- We typically represent vectors as column vectors

Not in Mortimer

# Matrix as Vector Operator

A vector $\vec{v}$ can by multiplied by a square matrix $\mathbf{A}$:

$$\vec{u} = \mathbf{A}\vec{v}$$

$$\begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$$

This can be seen as

$$\vec{v} \xrightarrow{\mathbf{A}} \vec{u}$$

# Properties of Matrix Multiplication

- Associative (결합법칙)

$$\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$$

- Distributive (분배법칙)

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$$

- Not necessarily commutative (교환법칙 X)

$$\mathbf{AB} \neq \mathbf{BA} \text{ (in some cases)}$$

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# What's Next: Full Bag of New Terms

1. Special matrices: identity matrix, zero matrix
2. Partner of a matrix: <u>inverse</u>, transpose, Hermitian conjugate
3. Properties of a matrix: singular, symmetric, Hermitian, orthogonal, unitary
4. Derived quantities from a matrix: trace, <u>determinant</u>

# Identity Matrix (항등행렬) $\mathbf{E}$

$$\mathbf{EA} = \mathbf{AE} = \mathbf{A}$$

$$\mathbf{E} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

$$E_{ij} = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

# Zero Matrix (영행렬) $\mathbf{0}$

$$\mathbf{0A} = \mathbf{A0} = \mathbf{0}$$

$$\mathbf{0} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

$$0_{ij} = 0$$

# Inverse of Matrix (역행렬)

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{E}$$

- Only square matrices have inverses
- In terms of matrix elements,

$$\sum_{k=1}^{n} a_{ik}(A^{-1})_{kj} = \delta_{ij}$$

# Finding Inverse: Gauss-Jordan Elimination

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{E}$$

- Apply a series of row operations on $\mathbf{A}$ and $\mathbf{E}$ to obtain

$$\mathbf{E}\mathbf{A}^{-1} = \mathbf{D}$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} (A^{-1})_{11} & (A^{-1})_{12} \\ (A^{-1})_{21} & (A^{-1})_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} (A^{-1})_{11} & (A^{-1})_{12} \\ (A^{-1})_{21} & (A^{-1})_{22} \end{bmatrix} = \begin{bmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{bmatrix}$$

# Row Operations

- Scalar Multiplication

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \rightarrow \begin{bmatrix} ca_{11} & ca_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

- Row subtraction

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \rightarrow \begin{bmatrix} a_{11} - a_{21} & a_{12} - a_{22} \\ a_{21} & a_{22} \end{bmatrix}$$

- Combination of the two

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \rightarrow \begin{bmatrix} a_{11} - ca_{21} & a_{12} - ca_{22} \\ a_{21} & a_{22} \end{bmatrix}$$

# Gauss-Jordan Elimination

- Construct an augmented matrix

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} (A^{-1})_{11} & (A^{-1})_{12} \\ (A^{-1})_{21} & (A^{-1})_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \left[\begin{array}{cc|cc} a_{11} & a_{12} & 1 & 0 \\ a_{21} & a_{22} & 0 & 1 \end{array}\right]$$

- Make an upper triangular matrix with diagonals = 1

$$\left[\begin{array}{cc|cc} 1 & * & * & * \\ 0 & 1 & * & * \end{array}\right]$$

- Then make the identity matrix

$$\left[\begin{array}{cc|cc} 1 & 0 & * & * \\ 0 & 1 & * & * \end{array}\right] \quad \mathbf{A}^{-1}$$

# Example 13.16

Find the inverse of the matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Construct an augmented matrix:

$$[\mathbf{A}|\mathbf{E}] = \left[\begin{array}{ccc|ccc} 2 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

Step 1: upper triangular matrix

Step 2: identity matrix

# Example 13.16

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

(1) Making upper triangular matrix

(Row 1) × 1/2:

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

(Row 2) – (Row 1):

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 3/2 & 1 & -1/2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

# Example 13.16

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 3/2 & 1 & -1/2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

(Row 2) × (2/3):

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 2/3 & -1/3 & 2/3 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

(Row 3) – (Row 2):

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 2/3 & -1/3 & 2/3 & 0 \\ 0 & 0 & 4/3 & 1/3 & -2/3 & 1 \end{array}\right]$$

# Example 13.16

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 2/3 & -1/3 & 2/3 & 0 \\ 0 & 0 & 4/3 & 1/3 & -2/3 & 1 \end{array}\right]$$

(Row 3) × (3/4):

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 2/3 & -1/3 & 2/3 & 0 \\ 0 & 0 & 1 & 1/4 & -1/2 & 3/4 \end{array}\right]$$

Now we have an upper triangular matrix with diagonals = 1.

# Example 13.16

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 2/3 & -1/3 & 2/3 & 0 \\ 0 & 0 & 1 & 1/4 & -1/2 & 3/4 \end{array}\right]$$

(2) Making identity matrix

(Row 2) – (2/3) × (Row 3):

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & -1/2 & 1 & -1/2 \\ 0 & 0 & 1 & 1/4 & -1/2 & 3/4 \end{array}\right]$$

(Row 1) – (1/2) × (Row 2):

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3/4 & -1/2 & 1/4 \\ 0 & 1 & 0 & -1/2 & 1 & -1/2 \\ 0 & 0 & 1 & 1/4 & -1/2 & 3/4 \end{array}\right]$$

# Example 13.16

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3/4 & -1/2 & 1/4 \\ 0 & 1 & 0 & -1/2 & 1 & -1/2 \\ 0 & 0 & 1 & 1/4 & -1/2 & 3/4 \end{array}\right]$$

Now, the inverse is

$$\mathbf{A}^{-1} = \begin{bmatrix} 3/4 & -1/2 & 1/4 \\ -1/2 & 1 & -1/2 \\ 1/4 & -1/2 & 3/4 \end{bmatrix}$$

Check

$$\mathbf{A}\mathbf{A}^{-1} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 3/4 & -1/2 & 1/4 \\ -1/2 & 1 & -1/2 \\ 1/4 & -1/2 & 3/4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

# Transpose of Matrix (전치행렬) $\tilde{\mathbf{A}}$

Rows → columns, columns → rows

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$$\tilde{\mathbf{A}} = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{bmatrix}$$

$$\left(\tilde{\mathbf{A}}\right)_{ij} = \tilde{a}_{ij} = a_{ji}$$

# Scalar Product Revisited

For two column vectors

$$\mathbf{V} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}, \qquad \mathbf{U} = \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix},$$

their scalar product can be seen as

$$\mathbf{V} \cdot \mathbf{U} = v_x u_x + v_y u_y + v_z u_z = \begin{bmatrix} v_x & v_y & v_z \end{bmatrix} \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix} = \tilde{\mathbf{V}}\mathbf{U}$$

# Hermitian Conjugate of Matrix $\mathbf{A}^\dagger$

## (에르미트 켤레 행렬)

Transpose + complex conjugate of each element

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$$\mathbf{A}^\dagger = \begin{bmatrix} a_{11}^* & a_{21}^* & a_{31}^* \\ a_{12}^* & a_{22}^* & a_{32}^* \end{bmatrix}$$

$$\left(\mathbf{A}^\dagger\right)_{ij} = a_{ji}^*$$

# Properties of a Square Matrix

- Singular matrix (특이행렬) has no inverse

- Symmetric matrix (대칭행렬): $\mathbf{A} = \widetilde{\mathbf{A}}$
- Hermitian matrix (에르미트행렬): $\mathbf{A} = \mathbf{A}^{\dagger}$

- Orthogonal matrix (직교행렬): $\mathbf{A}^{-1} = \widetilde{\mathbf{A}}$
- Unitary matrix (유니타리행렬): $\mathbf{A}^{-1} = \mathbf{A}^{\dagger}$

# Trace of a Square Matrix (대각합)

$$\operatorname{tr} \mathbf{A} = \sum_{i=1}^{n} a_{ii}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \rightarrow \operatorname{tr} \mathbf{A} = a_{11} + a_{22} + a_{33}$$

# Property of Trace

Even when $\mathbf{AB} \neq \mathbf{BA}$,

$$\mathrm{tr}(\mathbf{AB}) = \mathrm{tr}(\mathbf{BA})$$

$$\mathrm{tr}(\mathbf{AB}) = \sum_{i=1}^{n} (\mathbf{AB})_{ii} = \sum_{i=1}^{n} \sum_{k=1}^{n} a_{ik} b_{ki}$$

$$= \sum_{k=1}^{n} \sum_{i=1}^{n} b_{ki} a_{ik} = \sum_{k=1}^{n} (\mathbf{BA})_{kk} = \mathrm{tr}(\mathbf{BA})$$